

Kurmathi Sai Prakash Reddy

 saiprakash95666 |  LinkedIn |  saiprakash95666@gmail.com | +91 9700484763

SUMMARY

Software Engineer with 2.5 years of experience building secure, scalable, and cloud-native applications using React, .NET Framework, , C#, Node.js, SQL Server and AWS. Strong expertise in backend API development, microservices, and database-driven systems, with hands-on experience delivering production software in insurance, analytics, and hospitality domains. Solid academic and practical foundation in Machine Learning, Applied AI and LLMs, including integrating CNN-based models into web applications. Passionate about clean code, system reliability, and building user-centric products

WORK EXPERIENCE

Software Engineer, Rediantt LLC

Aug 2025 – Feb 2026

- Developed cloud-native Risk Analytics and Insurance Claims Optimization platform using C#, ASP.NET Core Web API, and SQL Server, processing thousands of transactions with 99.9% uptime.
- Built React dashboards with JavaScript (ES6+), React Hooks, and Redux for claims analysts and risk managers, improving workflow efficiency by 40%.
- Integrated Apache Kafka for event-driven microservices communication, enabling asynchronous processing and improved system scalability.
- Implemented AWS IAM, and encryption at rest to achieve HIPAA and SOX compliance across all data access layers.
- Built CI/CD pipelines using Docker, AWS CodePipeline, and CodeBuild, reducing deployment time by 35% and enabling daily releases.
- Managed cloud infrastructure with AWS ECS, Lambda, RDS, and CloudFormation (IaC) for reliable, version-controlled deployments.
- Implemented CloudWatch monitoring, logging, and alerting to proactively detect issues and reduce mean time to resolution by 40%.
- Developed unit and integration tests using xUnit and Moq with TDD approach, achieving 85% test coverage and reducing production defects.

Software Engineer, NCR Corporation (NCR Voyix)

Jan 2022 – Jul 2023

- Contributed to the modernization of the Aloha Cloud hospitality platform by delivering full-stack solutions across UI, API, and database layers.
- Migrated legacy Ext JS and .NET Framework to React with TypeScript, .NET 8+, and ASP .NET Web APIs improving performance by 50% and reducing page load times.
- Configured Webpack and Vite for optimized builds and managed frontend dependencies with npm, improving build times by 40%.
- Implemented comprehensive testing using Jest for unit tests and Playwright for E2E testing with TypeScript, achieving 80%+ coverage and reducing UI bugs by 60%.
- Designed RESTful APIs using ASP.NET Core Web API with Postman and OpenAPI/Swagger documentation to support third-party integrations.
- Optimized SQL Server queries and LINQ-based data access patterns, reducing data retrieval latency by 50% for high-traffic endpoints.
- Followed Agile/Scrum and TDD methodologies, participating in sprint planning, code reviews, root cause analysis, and zero-downtime production releases.

- Web Development Intern, The Sparks FoundationSep 2020 – Dec 2020
- Developed responsive web banking application using JavaScript, HTML5, and CSS3 with secure fund transfer and balance inquiry features.
 - Deployed application using GitHub Pages, achieving 100+ views within the first month and demonstrating end-to-end project ownership.
 - Managed source control using Git and GitHub with clean commit history and branching strategies for collaborative development.

MACHINE LEARNING & AI PROJECTS

- **Multi-Agent AI Bedtime Story Generator (LLMs, GPT-3.5)**
Designed and implemented a multi-agent AI system to generate age-appropriate bedtime stories for children aged 5–10. Orchestrated multiple autonomous agents with distinct roles (story planning, narrative generation, feedback evaluation, and refinement) using GPT-3.5-turbo. Built prompt pipelines and agent coordination logic to ensure coherent storytelling and safety-filtered output, demonstrating applied LLM system design, prompt engineering, and AI-driven content generation.
- **Emotion-Based Content Recommendation Web Application (CNN, FER2013)**
Designed and implemented an end-to-end web application that detects real-time facial emotions using a pre-trained CNN fine-tuned on the FER2013 dataset. Integrated camera-based image capture for facial emotion analysis and dynamically redirected users to personalized Spotify and YouTube content based on detected emotions. Documented model training, fine-tuning strategy, system architecture, and inference workflow to demonstrate applied ML-to-product integration.

EDUCATION

- 2023 – 2025Master of Science in Computer Science, **University of Central Missouri**
Coursework: Machine Learning, Deep Learning, NLP, Cloud Computing
- 2018 – 2022Bachelor of Technology in Computer Science, **KL University, India**
Specialization in Data Science and AI

SKILLS

Languages	C#, Java, TypeScript, JavaScript, SQL, Python, C++
Frontend	ReactJS, TypeScript, JavaScript (ES6+), Redux, React Hooks, HTML5, CSS3, Ext JS, Responsive Design, Webpack, Vite, npm, Telerik UI
Backend	Python (Flask, FastAPI), ASP.NET Core Web API, Node.js, REST APIs, Microservices, API Versioning, GraphQL, Distributed Systems
Databases	SQL Server, PostgreSQL, Amazon RDS, Relational & NoSQL Design, Query Optimization, Database Schema Design
Cloud	AWS(ECS, Lambda, RDS, CloudWatch), IAM, CI/CD
DevOps & Testing	Docker, CI/CD Pipelines, Automated Testing, Jest, Playwright, xUnit, Moq, Unit Testing, Integration Testing, Monitoring & Logging
LLM & AI Integration	GPT-3.5/4, Prompt Engineering, Multi-Agent Systems, LangChain, Model Fine-Tuning, Real-time AI Integration, RAG (Retrieval-Augmented Generation),ONNX Runtime
Machine Learning	Machine Learning, Deep Learning, CNNs, Computer Vision, Large Language Models (LLMs), Model Inference Integration, NumPy, Pandas
Practices	Agile/Scrum, Test-Driven Development (TDD), Clean Code, Code Reviews, Root Cause Analysis, Technical Documentation, System Design, Performance Optimization